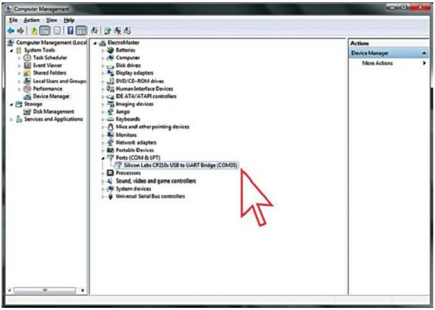

Fig. 3: Windows Device Manager

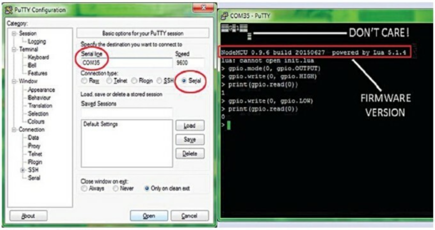

Fig. 4: Lua shell window

(1.6.9). Here is the code to change the onboard blue LED to blink like a breathing LED:

```
// Breathing LED on NodeMCU v1.0
   (Breathing.ino)
// Arduino IDE 1.6.9
// Based on a code published in
   arduining.com

#define LED D0 // Blue LED in NodeMCU at
   pin GPIO16 (D0)
#define BRIGHT 350 //max LED intensity
   (1-500)
#define INHALE 1250 //Inhalation time in
   milliseconds
#define PULSE INHALE*1000/BRIGHT
#define REST 1000 //Rest Between
   Inhalations
void setup() {
pinMode(LED, OUTPUT); // LED pin as
   output
}
void loop() {
//ramp increasing intensity, Inhalation
for (int i=1;i<BRIGHT;i++){
digitalWrite(LED, LOW); // turn the
   LED on
delayMicroseconds(i*10); // wait
digitalWrite(LED, HIGH); // turn the
   LED off
delayMicroseconds(PULSE-i*10); // wait
delay(0); // prevent watchdog firing
}
//ramp decreasing intensity, Exhalation
   (half time)
for (int i=BRIGHT-1;i>0;i--){
digitalWrite(LED, LOW); // turn the
   LED on
delayMicroseconds(i*10); // wait
digitalWrite(LED, HIGH); // turn the
   LED off
delayMicroseconds(PULSE-i*10); // wait
i--;
delay(0); // prevent watchdog firing
}
delay(REST); //take a rest
}
```

To program NodeMCU in Arduino IDE, install esp8266 board (ESP8266 core for Arduino) to Arduino IDE by following these simple steps:

1. Open Arduino IDE and select 'Preferences' via 'File' menu
2. Type http://arduino.esp8266.com/stable/package_esp8266com_index.json in the field for 'Additional Boards Manager URL'
3. Select 'Boards Manager' from 'Tools' menu
4. Under 'Boards Manager,' scroll until you find 'esp8266' board
5. Select the latest version and install
6. Select the board via 'Tools' menu (see Fig. 5)
7. Finally, copy, paste and then upload 'Breathing.ino' sketch

NodeMCU firmware flashing

Once NodeMCU is programmed through Arduino IDE, the original firmware will be erased. For some reason, if you want to restore the original firmware with Lua shell, you should flash the firmware again.

NodeMCU Flasher (*https://github.com/nodemcu/nodemcu-flasher*) is a popular firmware flash tool for Windows platform. After downloading and installing NodeMCU Flasher, download the latest NodeMCU firmware (*https://github.com/nodemcu/nodemcu-firmware/releases*).

At the time of writing this article, there are two versions for each firmware: integer (which supports only integer operations) and float (which supports floating-point cal-